JAVASCRIPT+ AI PROMPT

WHAT IS JAVASCRIPT, **WHY JAVASCRIPT** AND **HOW TO WRITE CODE USING JAVASCRIPT AND AI PROMT**.

AI tool that we are going to use?



Cursor IDE



ChatGPT

BY THE END OF THIS COURSE, YOU WILL BECOME A **PROMPT ENGINEER** AND **VIBE CODING ENGINEER** USING JAVASCRIPT.

TheLastCodeBender

PART 1: WHAT IS JAVASCRIPT?

Introduction to JavaScript

JavaScript is a **programming language** that makes your website interactive.
It runs in the **browser**, allowing you to do dynamic actions, something like:

- Show/hide elements (buttons, text, label, box etc.)
- Calculation
- Respond to clicks, hover, onKeyUp etc
- Validate forms
- Build full applications in the browser!
- You add JavaScript to HTML using the **<script>** tag or by linking a **.js** file.

PART 2: WRITING AND PRINTING TO THE CONSOLE

+ AI PROMPT

The console is a tool in the browser that helps us see outputs and debug JavaScript code.
We use the **console** object and the **log** method to print messages like this:

console.log(”message”);

There are two common ways to run JavaScript:

- From the command line using Node.js
- Directly in the browser using the developer console.

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM

- AI **Prompt:**

Generate a JavaScript program that prints "Welcome to JavaScript" in the console.

PART 3: VARIABLES AND DATA TYPES + AI PROMPT

Variables

We use variables to temporarily store data in our JavaScript programs.

In JavaScript, we use the **let, const, var** keywords to declare a variable.

- **let** – can change later when the program is running.
- **const** – stays constant
- **var** – older way

Keyword - variable name = value;

PART 3: VARIABLES AND DATA TYPES + AI PROMPT

Data Types

In JavaScript, data types define the kind of value a variable can hold, such as numbers, strings, booleans, objects, and more.

- **String**: "Hello"
- **Number**: 42
- **Boolean**: true, false
- **Array**, **Object**, **Null**, **Undefined**

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM- VARIABLES AND DATA TYPES

- AI **Prompt:**

Generate a JavaScript program that declares three variables: a string, a number, and a boolean.

Then print them to the console.

PART 4: OPERATORS + AI PROMPT

Operators in JavaScript are symbols used to perform operations on values, such as addition (+), comparison (==), or logical checks (&&). Operators let you do things with values such as:

- **Arithmetic**: +, -, *, /, %
- **Assignment**: =, +=, -=
- **Comparison**: ===, !==, <, >
- **Logical**: &&, ||, !

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM

☐ AI **Prompt:**

Generate a JavaScript program that performs arithmetic operations between two numbers and prints the results to the console.

- Then, generate another JavaScript program that does the following:
- Given a student score of 85, check if the student receives an A+, A, or B using comparison operators only. Don't use if statement, use comparison operators only.

PART 5: ARRAYS + AI PROMPT

An array in JavaScript is a special type of variable that can hold multiple values in a single list, organized by index.

Example of an array:

let fruits = ["Apple", "Banana", "Orange"];

Arrays are indexed starting from **0**.

The **length** of this array is **3**.

Print an array value:

console.log(**fruits**[2]); // Output: Orange

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM - ARRAY PROGRAM

☐ AI **Prompt:**

Generate a JavaScript program that creates an array of 5 colors and prints the second index in the console. Also, another array of pets and prints the last index in the console.

PART 6: OBJECTS + AI PROMPT

Objects in JavaScript help group information together using keys (also called properties) and their values.

Example of an object:

```
let person = {
  name: ”Ronaldo",
  age: 30,
 country: “USA”
};
console.log(person.name);
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – OBJECT PROGRAM

☐ AI **Prompt:**

Generate a JavaScript program that creates an array of 5 car objects. Each car should have make, model, and year as keys. Then, print the make of the first car and the model of the third car to the console.

PART 7: IF ELSE STATEMENT + AI PROMPT

In JavaScript, if-else statements let you run one piece of code if a condition is true and another if it's not.

Example:

```
let age = 20;
if (age > 18) {
  console.log("Adult");
} else {
  console.log("Minor");
}
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM - IF ELSE PROGRAM

- AI **Prompt:**

Generate a JavaScript program that checks if a number is positive, negative, or zero using if-else statements.

Another program to check and grade a student exam using if else statement given the student exam marks is 87%.

PART 8: SWITCH STATEMENTS + AI PROMPT

A switch statement in JavaScript is used to run different blocks of code based on the value of a variable or expression, as an alternative to multiple if-else statements.

Example:

```
const userRole = ”admin";

switch (userRole) {

  case ”admin":

    console.log(”Good morning, you have the privilege to use and control the entire system");

    break;

}
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – SWITCH STATEMENT PROGRAM

- AI **Prompt:**

Generate a JavaScript program using switch statement only that prints a message based on the day of the week.

PART 9: FOR LOOP + AI PROMPT

A for loop in JavaScript is used to repeatedly run a block of code a specific number of times, usually by using a counter variable.
Example:

```
for (let i = 1; i <= 5; i++) {
  console.log(i);
}
```

- **Initialization**: let i = 1
- **Condition**: i <= 5
- **Increment**: i++
- All in one line (inside the for statement)

We can use for loop to print all the values of an array.

Let see a practical example…..

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – FOR LOOP PROGRAM

- AI **Prompt:**

Generate a JavaScript program that print the values of array of fruits using for loop only.

Another program that print the values of 5 objects of cars using for loop only.

Note: don't use any method, use for loop only, keep it simple.

PART 10: WHILE LOOP + AI PROMPT

A while loop in JavaScript repeats a block of code as long as a specified condition remains true. A **while loop** is very similar to a **for loop**. In a **while loop**, we do the **initialization**, **condition check**, and **increment** in separate places. Whereas in a **for loop**, the **initialization**, **condition**, and **increment** are all done in one place within the loop declaration. Example:

```
let i = 0;   //Initialization

while (i < 5) {  //Condition

  console.log(i);

  i++;    //Increment

}
```

- Initialization, condition, and increment are done in **separate places**
- Same output as the for loop

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – WHILE LOOP PROGRAM

- AI **Prompt:**

Generate a JavaScript program that prints even numbers from 0 to 10 using while loop only.

PART 11: FOREACH LOOP + AI PROMPT

The forEach method in JavaScript is used to loop through each item in an array and run a specified function on each element.

Example:

```
let fruits = ["Apple", "Banana”,” Orange”];

fruits.forEach(function (fruit) {

  console.log(fruit);

});
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM FOREACH LOOP PROGRAM

- AI **Prompt:**

Generate a JavaScript program with array of 4 animal names and use forEach to print each one.

PART 12: MAP + AI PROMPT

JavaScript's map lets you change every item in an array and gives you a new array with the results.

Example:

```
let numbers = [1, 2, 3];

let addTwoToEachNumber = numbers.map(n => n + 2);
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – MAP PROGRAM

- AI **Prompt:**

Generate a JavaScript program with an array of numbers and use map() to double each number, and plus 10 then print the result.

PART 13: METHODS/FUNCTIONS + AI PROMPT

Methods or Functions in JavaScript are reusable blocks of code designed to perform specific tasks.

There are a few ways to define a function: eg,

- Function Declaration
- Function with Parameters
- Arrow Functions

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM – FUNCTION PROGRAM

- AI **Prompt:**

Generate a JavaScript program that defines a function called addNumbers which takes two numbers as parameters, returns their sum, and prints the result.

Another function that take student marks and check if the student receives an A+, A, B, C, D or F using switch statement and return the result.

PART 14: READING VALUES FROM USER INPUT AND PRINTING TO THE CONSOLE + AI PROMPT

In JavaScript, we can **read input from the user** using the prompt() function in the browser.

It's a simple built-in way to get text input.

Example:

```
let name = prompt("What is your name?");

console.log("Hello, " + name + "!");
```

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM READING USER INPUTS PROGRAM

- AI **Prompt:**

Generate a JavaScript program that ask and reads input from the user, the program should ask user information and then displays a personalized message.

Another program that will ask and read user information and check if the user is qualify to drive a car or not

PART 15: EXCEPTION HANDLING + AI PROMPT

Exception handling in JavaScript is the process of detecting and managing errors that occur during program execution, so the program doesn't crash unexpectedly. It's done using **try...catch blocks.**

Why We Need It

- Prevents the program from stopping when an error occurs.
- Allows us to handle errors and show friendly messages to users.
- Helps in debugging by logging error details.
- Improves program stability and user experience.

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM EXCEPTION HANDLING PROGRAM

☐ AI **Prompt:**

Create a JavaScript program that demonstrates exception handling using try...catch.

- The program should ask the user to enter a number.
- If the user enters a non-numeric value, throw an error with the message "Invalid number entered".
- Catch the error and display it using console.error().
- If the input is valid, print "You entered a valid number"

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM PROJECT ONE – ADMISSION INTO THE UNIVERSITY ELIGIBILITY CHECKER WITH PROMT() FUNCTION

- AI **Prompt:**

Generate a JavaScript program that asks for and reads user information, then checks if the user is eligible for admission into the university (Software Engineering Program).

AI PROMPT TO GENERATE JAVASCRIPT PROGRAM PROJECT TWO – BUILDING A CALCULATOR EXACTLY LIKE IPHONE CALCULATOR USING HTM,CSS AND JAVASCRIPT

☐ AI **Prompt:**

Prompt One:

Create a CALCULATOR Exactly like iPhone Calculator using HTM,CSS.

- Create a separate file for the CSS .
- Make it very beautiful and user friendly.

NOTE: Don't do any JavaScript yet, focus on the user interface first. I will

Prompt Two:

Use JavaScript and complete the Calculator, make it function using JavaScript.

- Keep the code simple and clean, and use exception handling to catch any error

THE END